**Capstone Second Review**

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|---|---|
| **Project Title** | **:** Privacy-Preserving Federated Learning for Missing Person Identification |
| **Domain** | **:** Computer Vision, Federated Learning & Biometric Privacy |
| **Reg. No** | **: 1.** 23BCE9603 **2.** 23BCE9585 **3.** 23BCE9746 **4.** 23BCE8540 |
| **Project Guide** | **:** Kanhaiya Sharma (**70880**) |
| **Academic Year** | **:** Fall Semester 2026–2027 |

## 1. Problem and Objectives:

Centralised CCTV face recognition requires pooling footage, which exposes bystanders and creates a single point of failure. FedMPS lets camera nodes improve a missing-person search together without sharing raw video, face crops or unprotected biometrics.

- **Objective 1:** multi-camera pipeline (detection, tracking, quality gate, 5-point alignment, ArcFace embeddings) trained by federated learning.
- **Objective 2:** pairwise-masked secure aggregation, differential privacy on updates and cancelable watchlist templates, each measured for accuracy cost.

## 2. System Architecture

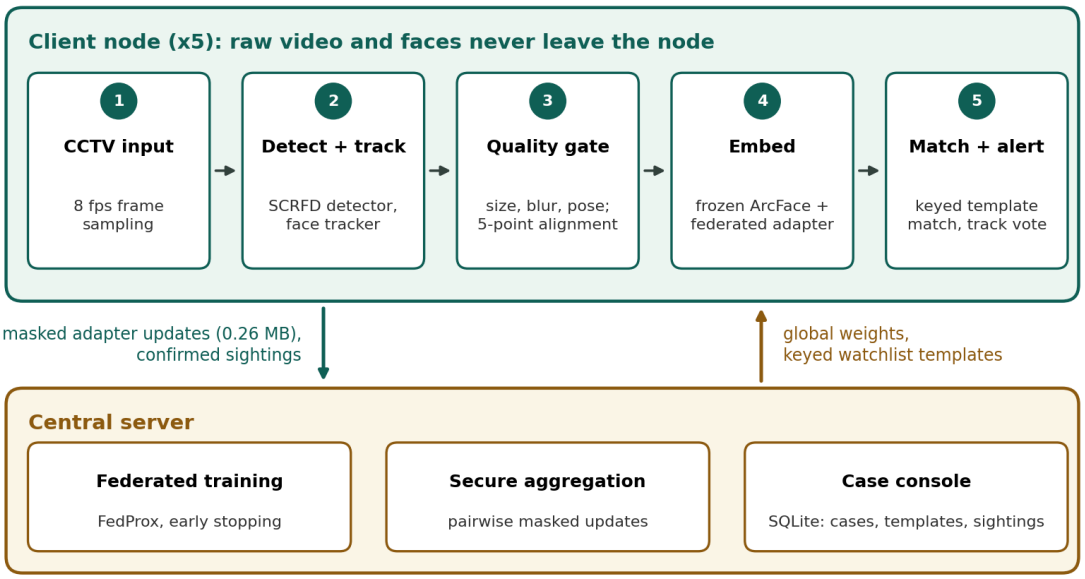



***Figure 1:*** *Five client nodes and one central server.*

The 43.6 M-parameter ArcFace backbone is frozen. Nodes train only a residual adapter (512-64-512, 66,240 parameters). One update is 0.26 MB, against about 174 MB for the full backbone (658x smaller).

## 3. Dataset and Partition:

***Table 1:*** *Dataset statistics: published VGGFace2 against the subset used here.*

| Statistic | VGGFace2 (original) | This project | Source |
|---|---|---|---|
| Subjects | 9,131 | 8,631 | Pipeline log |
| Images | 3.31 million | 172,620 | Pipeline log |
| Images per subject (min/avg/max) | 80 / 362.6 / 843 | 20 / 20 / 20 | Pipeline log |
| Landmark alignment | 5 landmarks provided | Re-aligned with det_10g; 14 missed (0.01%) | Pipeline log |

*Original statistics: Cao et al., VGGFace2, 2018. Our subset is the 8,631-identity training split.*

Data is split by identity. 70% of identities (6,042; 120,840 images) are held out for testing and never used in training. The other 30% (2,589 identities) is split unevenly across five nodes, each with its own degradation profile (face size 20-48 px, blur 1-9 px, noise up to 12, gamma 0.7-1.6, JPEG quality 25-70).

***Table 2:*** *Identity and image counts per federated node.*

| Node | Identities | Train images | Val images | Total |
|---|---|---|---|---|
| Node 0 | 258 | 4,120 | 1,040 | 5,160 |
| Node 1 | 388 | 6,200 | 1,560 | 7,760 |
| Node 2 | 517 | 8,280 | 2,060 | 10,340 |
| Node 3 | 647 | 10,360 | 2,580 | 12,940 |
| Node 4 | 779 | 12,460 | 3,120 | 15,580 |

*Train and validation are split by identity, not by image: 20% of each node's identities (52, 78, 103, 129 and 156 respectively) are held out of training entirely, so early stopping is measured on people that node never trained on. Training runs 30 federated rounds of one local epoch each, so 30 epochs per node and 150 node-epochs in total, at batch size 128. One round passes over 41,420 images across the five nodes; 1,242,600 image-passes over the full run.*

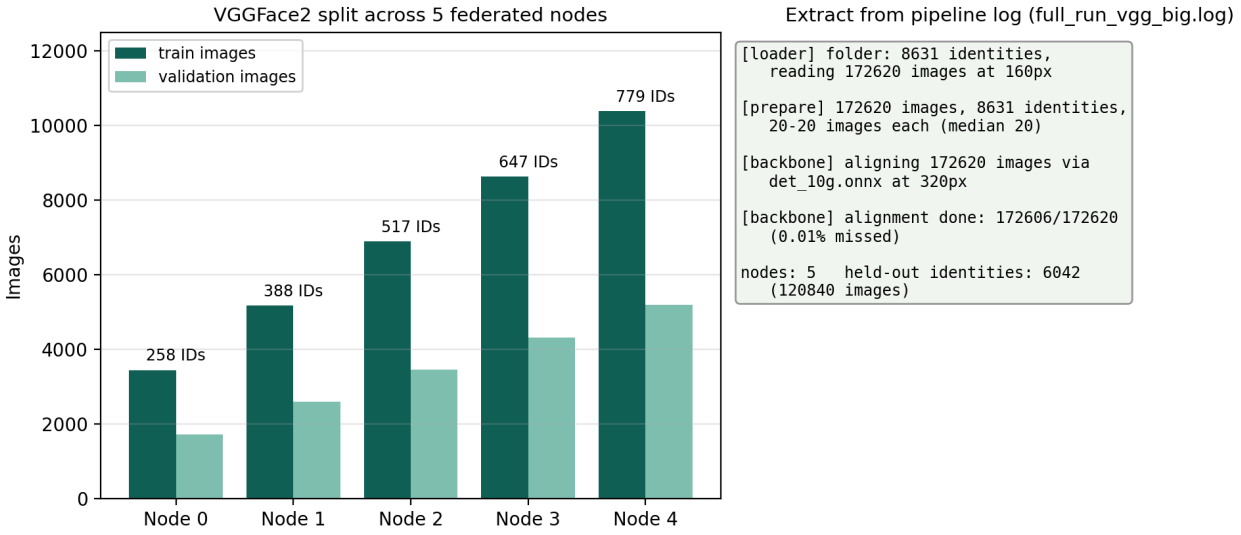



***Figure 2:*** *Node split and pipeline log extract.*

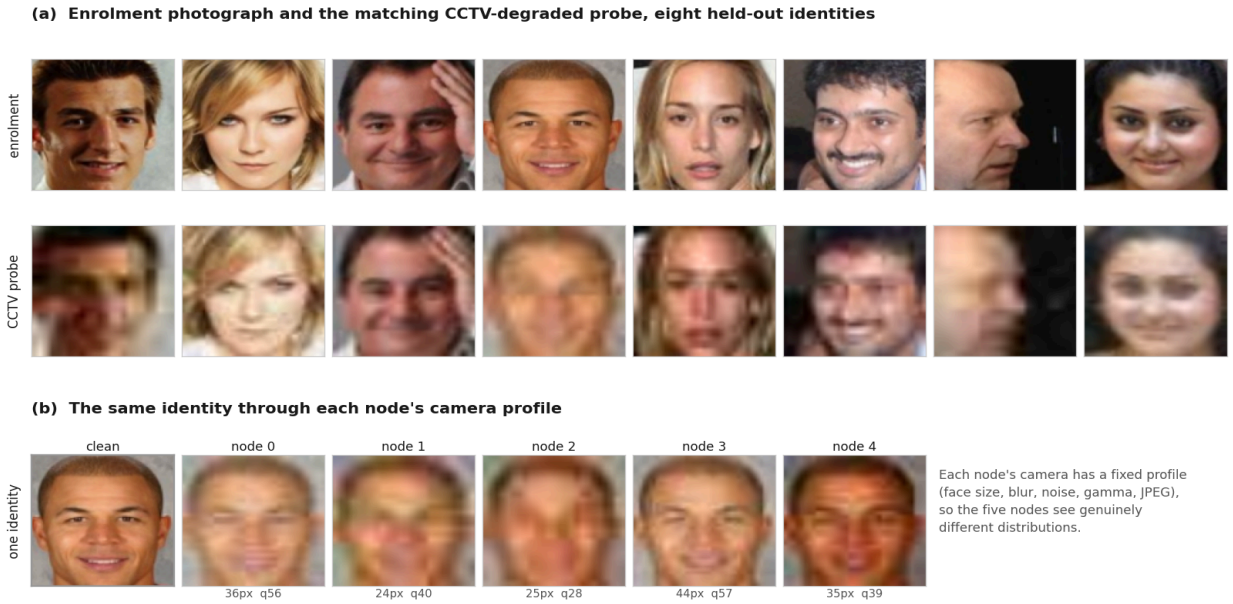



***Figure 3:*** *Aligned 112x112 crops as the backbone sees them.Degradation is synthetic;*

## 4. Test Protocol:

***Table 3:*** *Evaluation protocol: gallery, probes, scoring rule and validation.*

| Item | Setting |
|---|---|
| Backbone | Pretrained ArcFace ResNet-50 (buffalo_l), frozen; not trained on our split. |
| Watchlist | 3,021 held-out identities, 3 clean photos each (averaged). |
| Non-mate crowd | 3,021 other held-out identities, never enrolled: 18,126 probes. |
| Mate probes | 15,105 degraded probes of enrolled people, each fused over 3 frames. Enrolment photos are never reused as probes. |
| Success rule | Score above threshold and correct top-1 identity. |
| Verification | 6,000 same/different pairs, clean vs degraded. |
| Fixed test set | One set (seed 1337), identical for every model. |
| Validation | Early stopping uses separate validation identities (20% of each node), not the test set. |

**Metrics.** Rank-1: enrolled person named first. TPIR @ FPIR x: enrolled people correctly flagged when a fraction x of non-enrolled people trigger a false alarm. TAR @ FAR x: the 1:1 verification equivalent. AUC and EER summarise the verification curve.

***Table 4:*** *Comparison arms and the question each one answers.*

| Compared against | Role | Purpose |
|---|---|---|
| Clean probes | Upper bound | Size of the CCTV degradation problem |
| Frozen backbone | Baseline | Whether the adapter adds anything |
| Local only (each node) | Lower bound | Whether federation helps |
| Centralised (pooled data) | Upper bound | Accuracy cost of keeping data local |

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## 5. Results:

### 5.1. Alignment and dataset:

Alignment A/B test (same 5,000 identities, seed and degradation): same-person cosine 0.292 to 0.514; cross-person 0.103 to 0.004. At 3 photos x 3 frames, rank-1 rose from 0.206 to 0.985. The first CelebA run used pre-aligned crops and hid this defect.

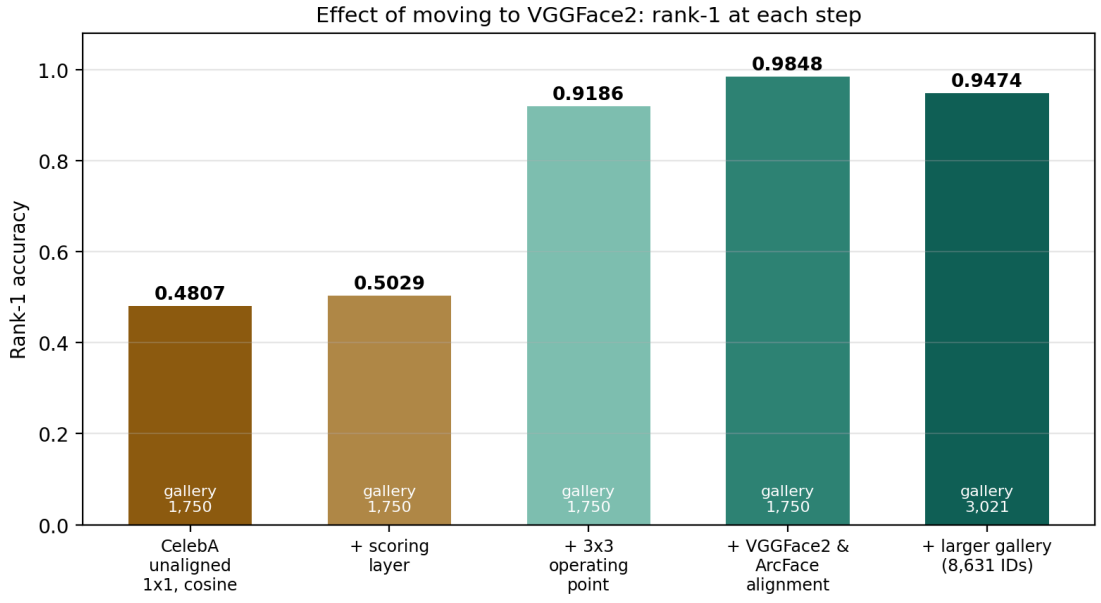



***Figure 4:*** *Rank-1 at each step. The scoring layer, 3x3 operating point, VGGFace2 and alignment were changed together, so the dataset effect cannot be isolated. The last step enlarges the gallery to 3,021 on purpose.*

### 5.2. Federated vs local vs centralised

***Table 5:*** *Verification and open-set identification by training regime.*

| Training regime | TAR @ FAR 0.1% | TAR @ FAR 1% | TPIR @ FPIR 0.1% | TPIR @ FPIR 1% | Rank-1 |
|---|---|---|---|---|---|
| Clean probes* | 0.9247 | 0.9347 | 0.9801 | 0.9915 | 0.9962 |
| Frozen backbone | 0.6287 | 0.7440 | 0.6725 | 0.7968 | 0.9474 |
| Centralised | 0.6040 | 0.7377 | 0.6830 | 0.7983 | 0.9452 |
| Federated (FedProx, 30 rounds) | 0.6180 | 0.7403 | 0.6821 | 0.8044 | 0.9448 |
| Local only (mean +/- std) | 0.564 +/- 0.022 | 0.715 +/- 0.017 | 0.637 +/- 0.028 | 0.770 +/- 0.016 | 0.9285 +/- 0.008 |

** Raw cosine scoring; other rows use the scoring layer. FAR below 0.1% is not reported (6,000 pairs cannot resolve it). Rank-5 and rank-10 for the frozen backbone are 0.9752 and 0.9814, recomputed from the saved embeddings on the same fixed test set. The federated adapter checkpoint was overwritten after the run, so its rank-5/10 could not be recovered and are left unreported.*

- Federated vs local-only: +1.63 points rank-1, +3.4 points TPIR @ 1% FPIR.
- Federated vs centralised: rank-1 gap 0.0004, TPIR @ 1% FPIR higher by 0.006. Single seed, no confidence intervals: treat as equal.
- Adapter vs frozen backbone: rank-1 lower by 0.0026, TPIR @ 1% FPIR higher by 0.008. No clear gain.

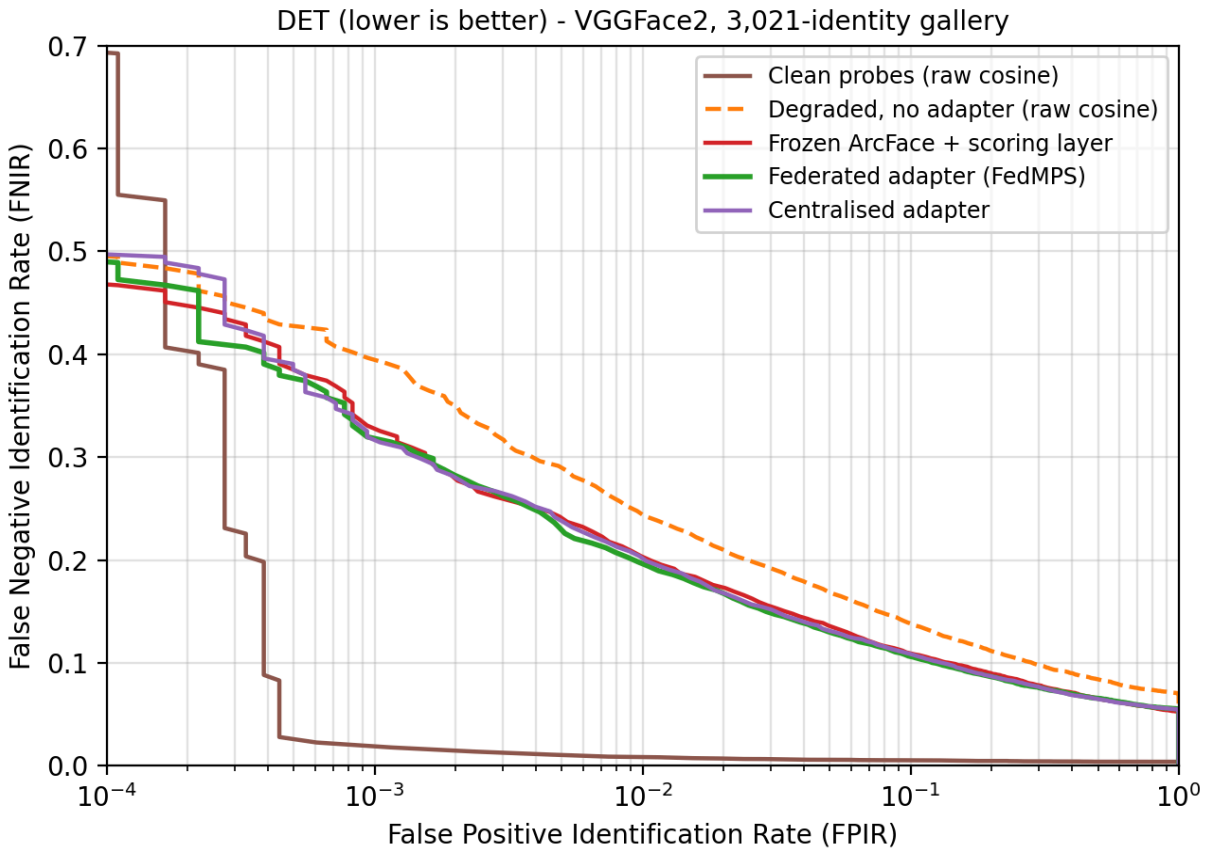



***Figure 5:*** *DET curves, 3,021-identity gallery. Clean and degraded use raw cosine; the other three use the scoring layer.*

***Table 6:*** *Verification summary: AUC, EER and TAR at fixed FAR.*

| Model | AUC | EER | TAR @ FAR 1% | TAR @ FAR 0.1% |
|---|---|---|---|---|
| Clean probes | 0.9769 | 0.050 | 0.9347 | 0.9247 |
| Frozen backbone | 0.9456 | 0.121 | 0.7440 | 0.6287 |
| Centralised | 0.9440 | 0.118 | 0.7377 | 0.6040 |
| Federated | 0.9448 | 0.119 | 0.7403 | 0.6180 |
| Local only (mean) | 0.9398 | 0.124 | 0.715 | 0.564 |

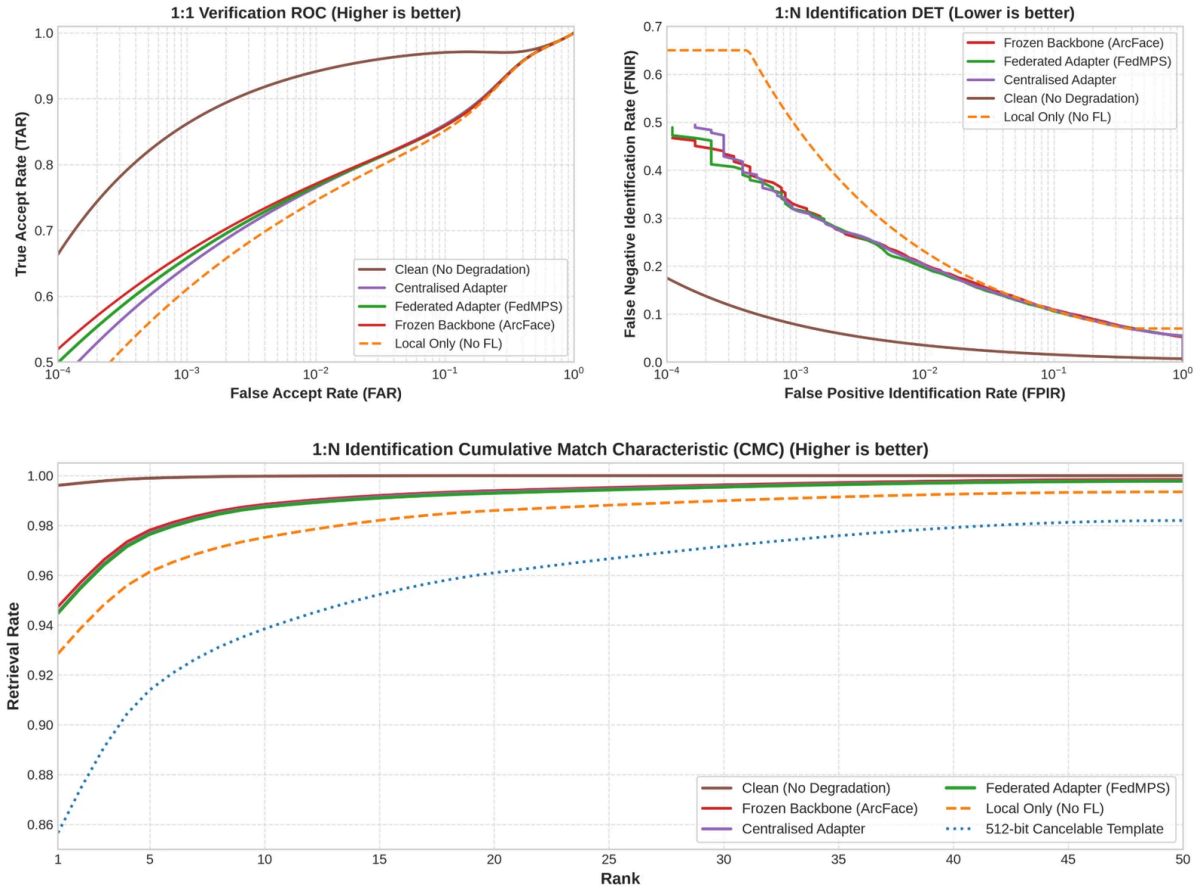



***Figure 6:*** *Verification ROC, open-set identification DET and closed-set CMC for all five arms on the 3,021-identity gallery. Rank-1 reads off the CMC at rank 1; the clean curve is the ceiling the degradation model costs.*

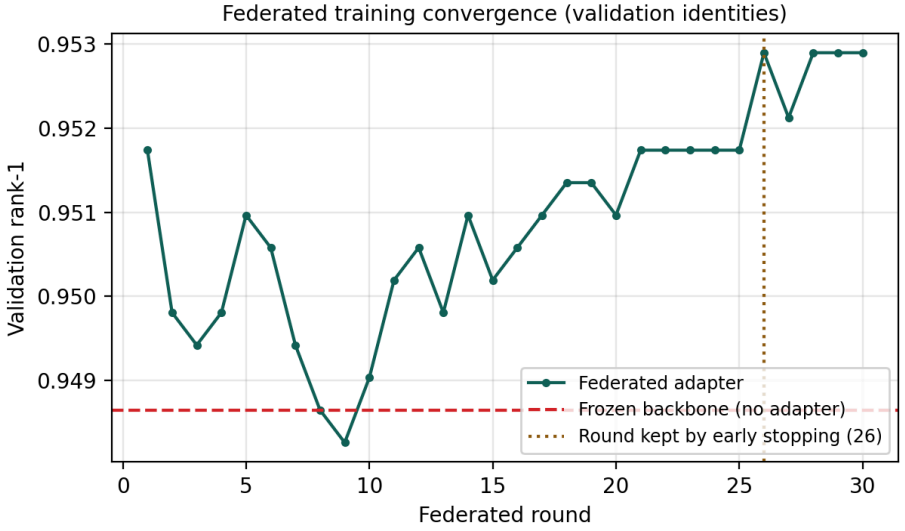



***Figure 7:*** *Validation rank-1 per round. Round 26 is kept (validation rank-1 0.9529 vs 0.9486 for the frozen backbone).*

### 5.3. Per-camera results:

***Table 7:*** *Per-camera rank-1 and TPIR, frozen backbone against federated adapter.*

| Camera | Frozen rank-1 | Federated rank-1 | Frozen TPIR @ 1% | Federated TPIR @ 1% | Gain @ 1% |
|---|---|---|---|---|---|
| Node 0 | 0.9539 | 0.9503 | 0.846 | 0.841 | -0.005 |
| Node 1 | 0.9200 | 0.9146 | 0.727 | 0.735 | +0.008 |
| Node 2 | 0.9234 | 0.9221 | 0.737 | 0.744 | +0.007 |
| Node 3 | 0.9782 | 0.9766 | 0.871 | 0.868 | -0.003 |
| Node 4 | 0.9608 | 0.9597 | 0.833 | 0.827 | -0.006 |

### 5.4. Photos and frames per case:

***Table 8:*** *Rank-1 by enrolment photographs and fused probe frames.*

| Enrolment photos x probe frames | Raw cosine | Full scoring layer | Gain |
|---|---|---|---|
| 1 x 1 | 0.5240 | 0.5419 | +0.018 |
| 1 x 3 | 0.8238 | 0.8511 | +0.027 |
| 3 x 3 (default) | 0.9294 | 0.9474 | +0.018 |
| 5 x 5 | 0.9861 | 0.9918 | +0.006 |

*Rank-1. The scoring layer is centring, rotation and hubness correction.*

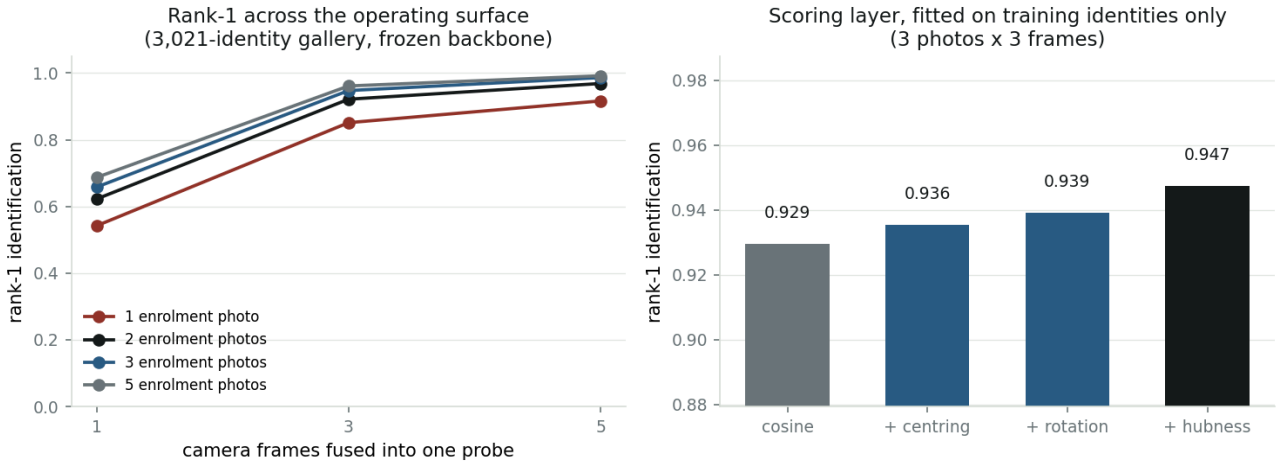



***Figure 8:*** *Rank-1 across enrolment and frame settings.*

### 5.5. Published VGGFace2 results (context):

***Table 9:*** *Published VGGFace2 results, shown for metric format only.*

| Model / training data | Test set | TAR @ FAR 0.1% | TPIR @ FPIR 1% | Rank-1 | Rank-5 |
|---|---|---|---|---|---|
| VGGFace, ResNet-50 [paper] | IJB-A | 0.620 | 0.454 | 0.925 | 0.972 |
| MS-Celeb-1M, ResNet-50 [paper] | IJB-A | 0.851 | 0.807 | 0.961 | 0.982 |
| VGGFace2, ResNet-50 [paper] | IJB-A | 0.895 | 0.844 | 0.976 | 0.992 |
| VGGFace2, ResNet-50 [paper] | IJB-B | 0.878 | 0.701 | 0.886 | 0.936 |
| VGGFace2, ResNet-50 [paper] | IJB-C | 0.900 | 0.735 | 0.898 | 0.939 |

| Model / training data | Test set | TAR @ FAR 0.1% | TPIR @ FPIR 1% | Rank-1 | Rank-5 |
|---|---|---|---|---|---|
| FedMPS federated (ours) | VGGFace2, CCTV-degraded | 0.618 | 0.804 | 0.945 | not saved |
| Frozen ArcFace (ours) | VGGFace2, CCTV-degraded | 0.629 | 0.797 | 0.947 | 0.975 |

*Different models, test sets and protocols (paper: 10-split IJB benchmarks, trained on VGGFace2; ours: pretrained model, degraded VGGFace2, one fixed split). Shown for metric format only; not comparable.*

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## 6. Privacy Evaluation:

***Table 10:*** *Privacy mechanisms: measured effect and cost.*

| Mechanism | Result | Cost / limit |
|---|---|---|
| Secure aggregation (pairwise masks, Diffie-Hellman 2048-bit) | Rank-1 0.9376 plain and masked (difference 0.0000). Masked update norm 513 vs 2.6 raw. | 1.008x run time (26.8 s to 27.0 s). No dropout recovery. |
| Differential privacy (clip 0.128, Gaussian noise) | Sigma 0: 0.9391; 0.3: 0.8751; 0.6: 0.6254; 1.0: 0.2477; 2.0: 0.0216. | Epsilon 252 at sigma 0.3, 16 at sigma 2.0. No setting gives epsilon < 10 with usable accuracy. |
| Membership inference (loss threshold) | AUC 0.4963 (chance 0.5). | One attack type only. |
| Cancelable templates (keyed binary, XOR) | Rank-1 0.9226 raw float, 0.8321 at 512 bits (64 bytes), 0.6943 at 256 bits, 0.3962 at 96 bits. | Costs about 9 points at 512 bits. Protects stored templates, not a compromised live node. |

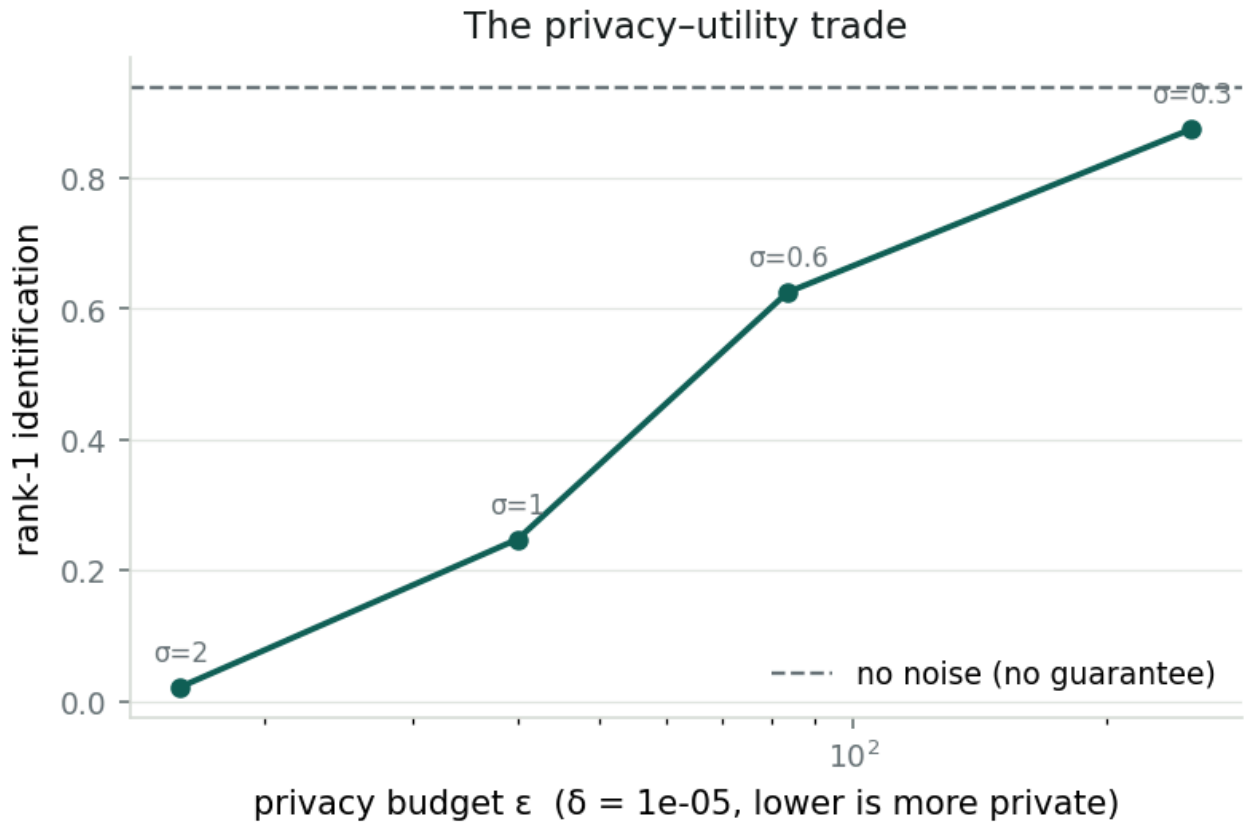



***Figure 9:*** *Differential privacy: noise vs accuracy.*

*Template figures come from a separate sweep on a smaller test set; absolute values differ from Section 5.*

## 7. Pipeline and Dashboards:

Server and node dashboards (Streamlit) share one SQLite store with an audit log. The server issues a keyed template per node. Each node detects, tracks, quality-gates and matches locally; a match becomes a PENDING alert on the node. Only alerts confirmed by a human reviewer reach the server. Cases expire, and closing a case revokes its templates.

Run on five 120-frame node clips (40 frames sampled): alerts on nodes 0, 2, 3 and 4, none on node 1; 0.43 to 2.8 frames per second.

## 8. Limitations:

- Degradation is synthetic; no real CCTV footage was tested.
- Single seed, no confidence intervals.
- Overlap between the pretrained backbone's training identities and VGGFace2 was not checked.
- Probes are fused over 3 frames; single-frame rank-1 is 0.54.
- The live video matcher uses raw cosine; the scoring layer is not yet integrated.
- Adapter shows no clear gain over the frozen backbone; differential privacy has no usable operating point.

## 9. Contributions:

***Table 11:*** *Deliverables by team member.*

| Member | Deliverables |
|---|---|
| G N Lokesh | VGGFace2 loader, landmark alignment, CCTV degradation, open-set evaluation metrics |
| R Kumaresh | Federated training loop, FedProx, three-way comparison, early stopping |
| K Kishore | Pairwise masking, differential privacy sweep, membership inference, cancelable templates |
| Aswin Maheswaran | Server and node dashboards, tracking, SQLite store and audit log |

## 10. Plan: Weeks 10-12

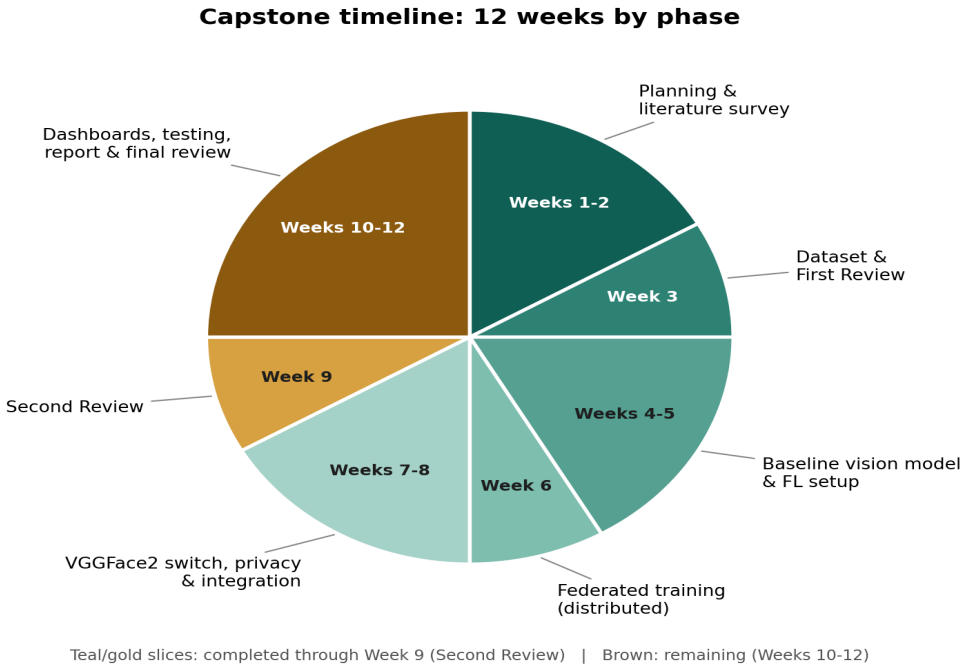



*Figure 10. Timeline by phase.*

- **Week 10:** integrate the scoring layer into the video matcher; finalise dashboards.
- **Week 11:** end-to-end testing, throughput and memory benchmark, draft report.
- **Week 12:** final report, slides, demonstration.